Distributed Optimization for Machine Learning

Lecture 21 – Analog computing for energy-efficient AI: Part I

ECE 5290/7290 & ORIE 5290

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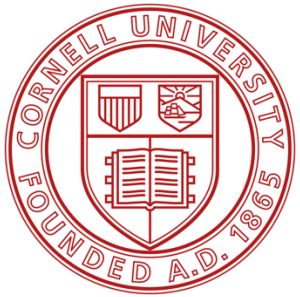
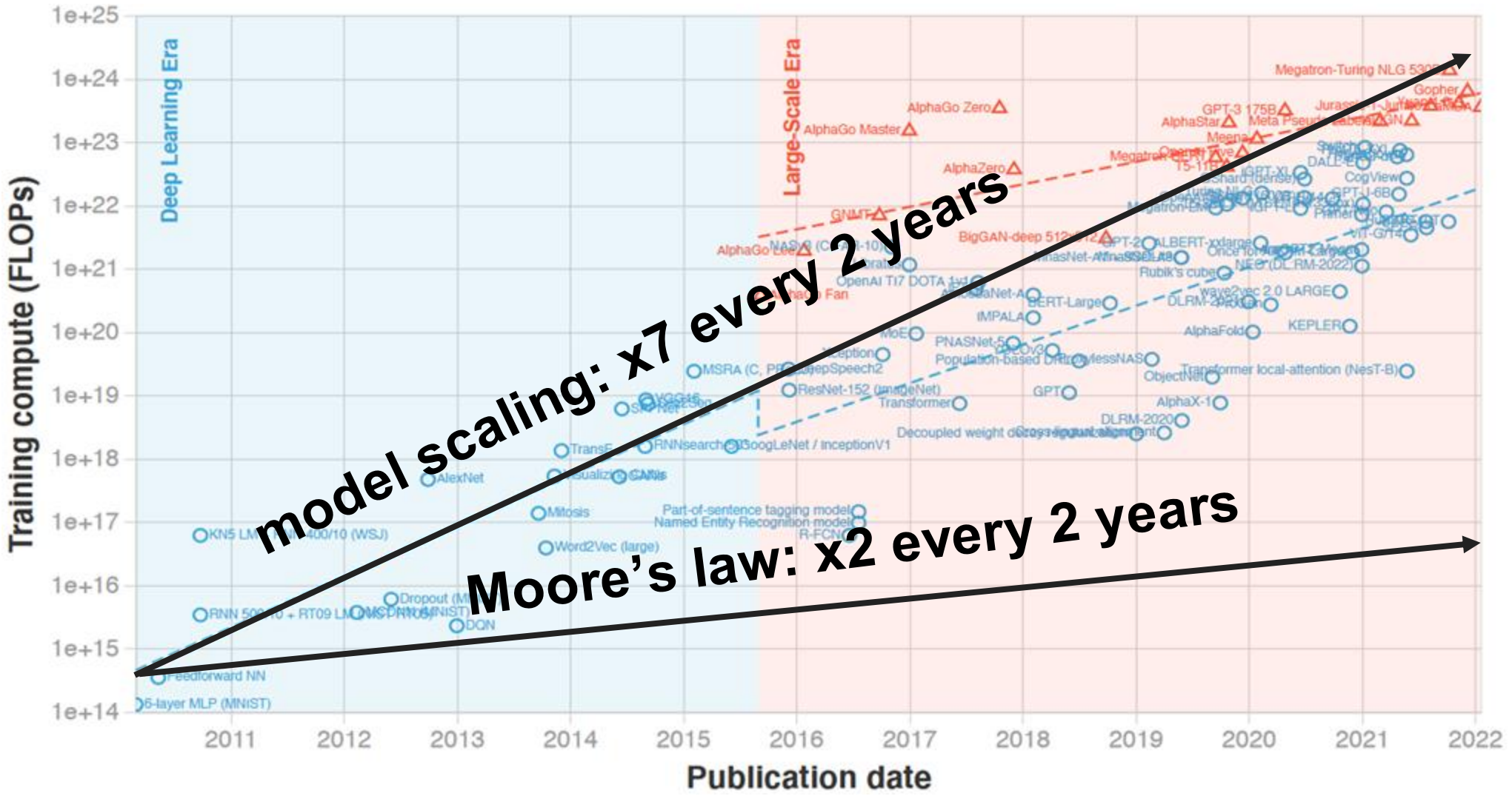
November 17, 2025

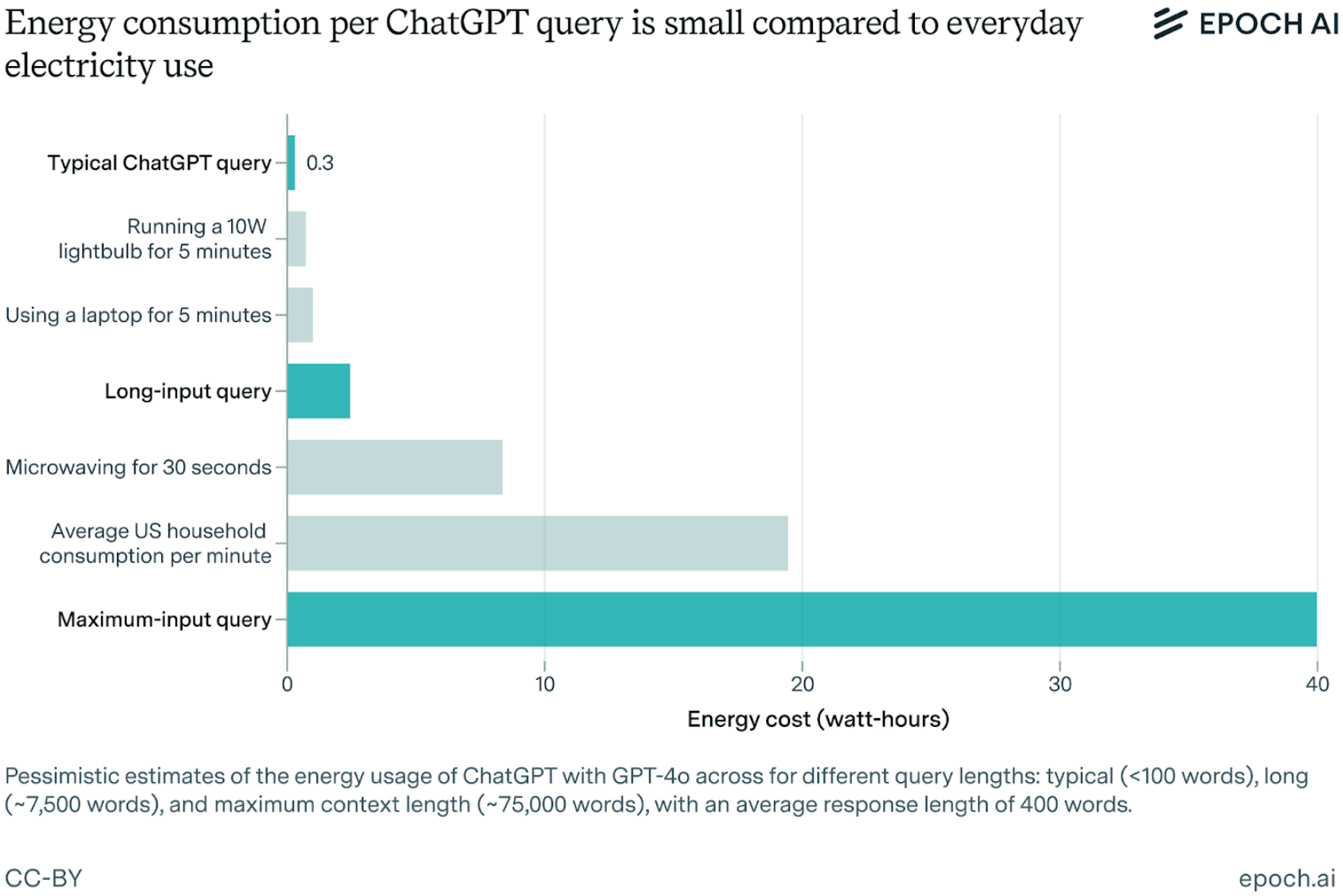


Training large models requires large compute



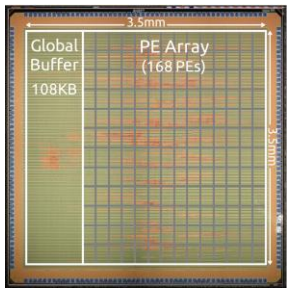
Sevilla et al., "Compute trends across three eras of machine learning," **IJCNN 2022.**

The carbon footprint of using ChatGPT



Source: Josh You at Epoch AI (How much energy does ChatGPT use?)

Current computing hardware



CPU **GPU** **FPGA** **ASIC**

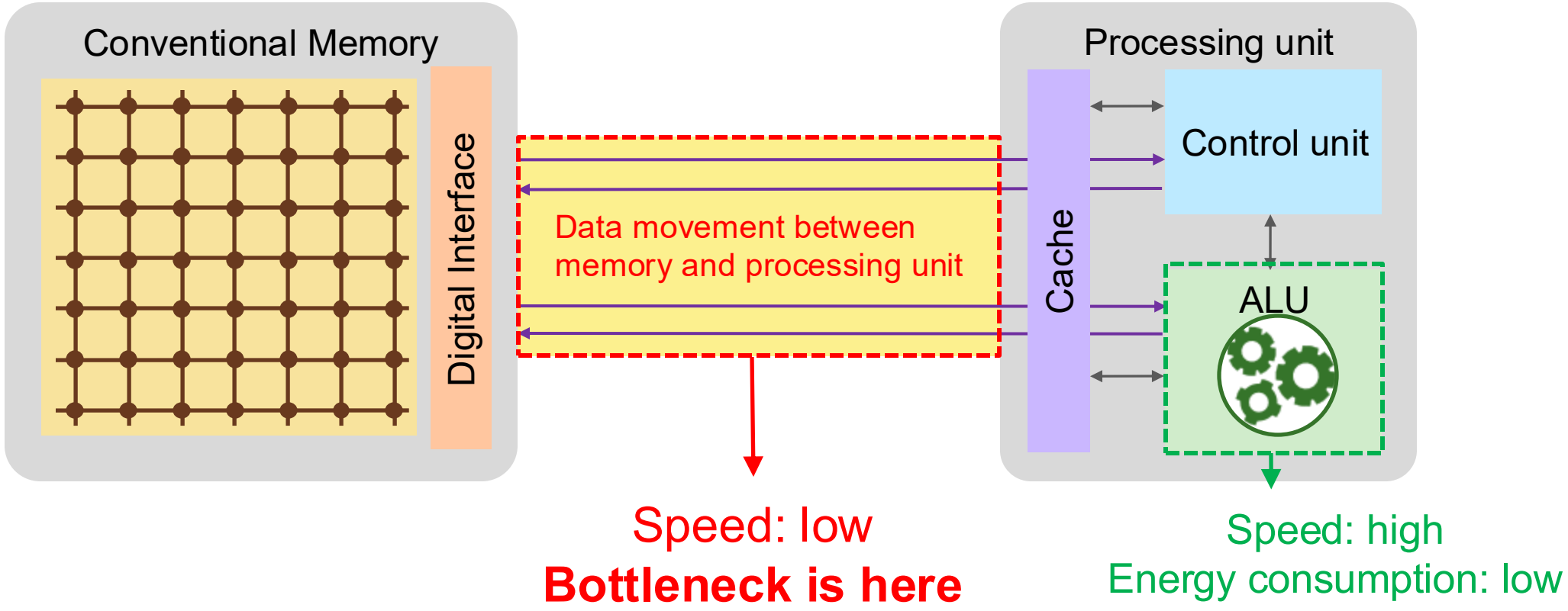
Flexibility **Efficiency**

They all belong to **out-memory computation architecture**

https://www.intel.com/content/www/us/en/newsroom/news/12th-gen-intel-core-i9-12900ks-worlds-fastest-desktop-processor.html#gs.9yghxm
https://www.nvidia.com/en-us/data-center/a100/
https://www.xilinx.com/products/boards-and-kits/1-t5lnks.html

Bottleneck: Data movement is costly

Out-memory architecture

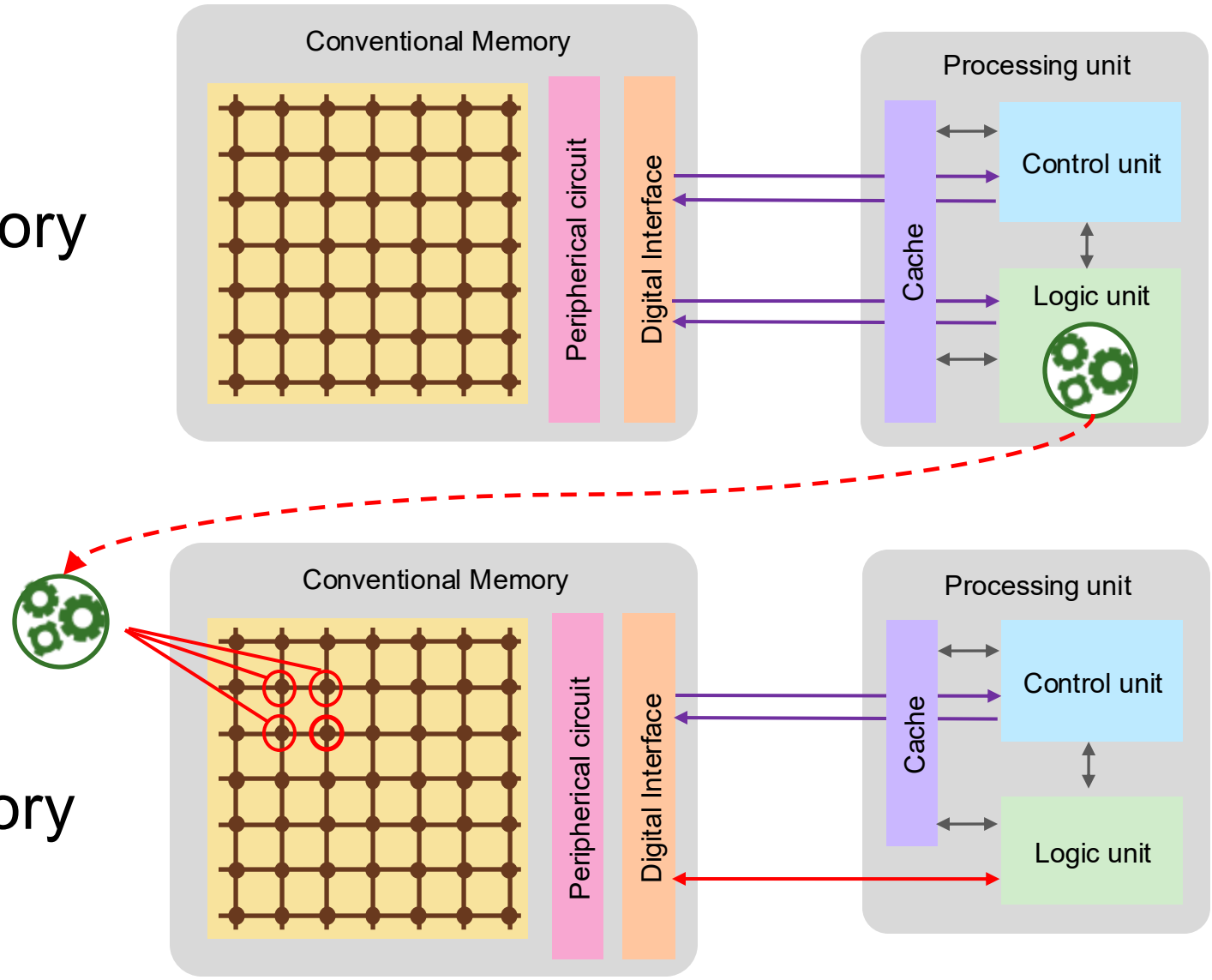


Out-memory vs in-memory computation

Out-memory

In-memory computation can be done via either **digital** or **analog** devices

In-memory



Data flow in deep models

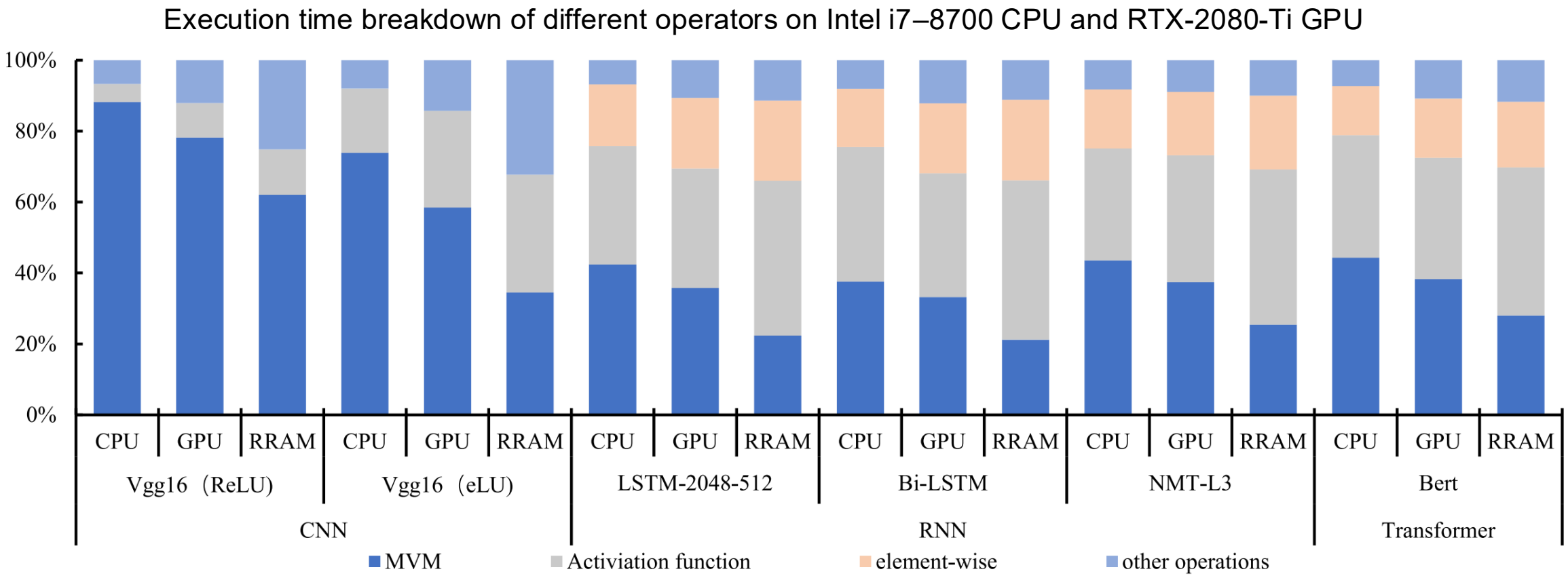
Linear operation
(Large matrix)

Non-linear activation
(Small vector)

Fix computationally intensive layers in memory!

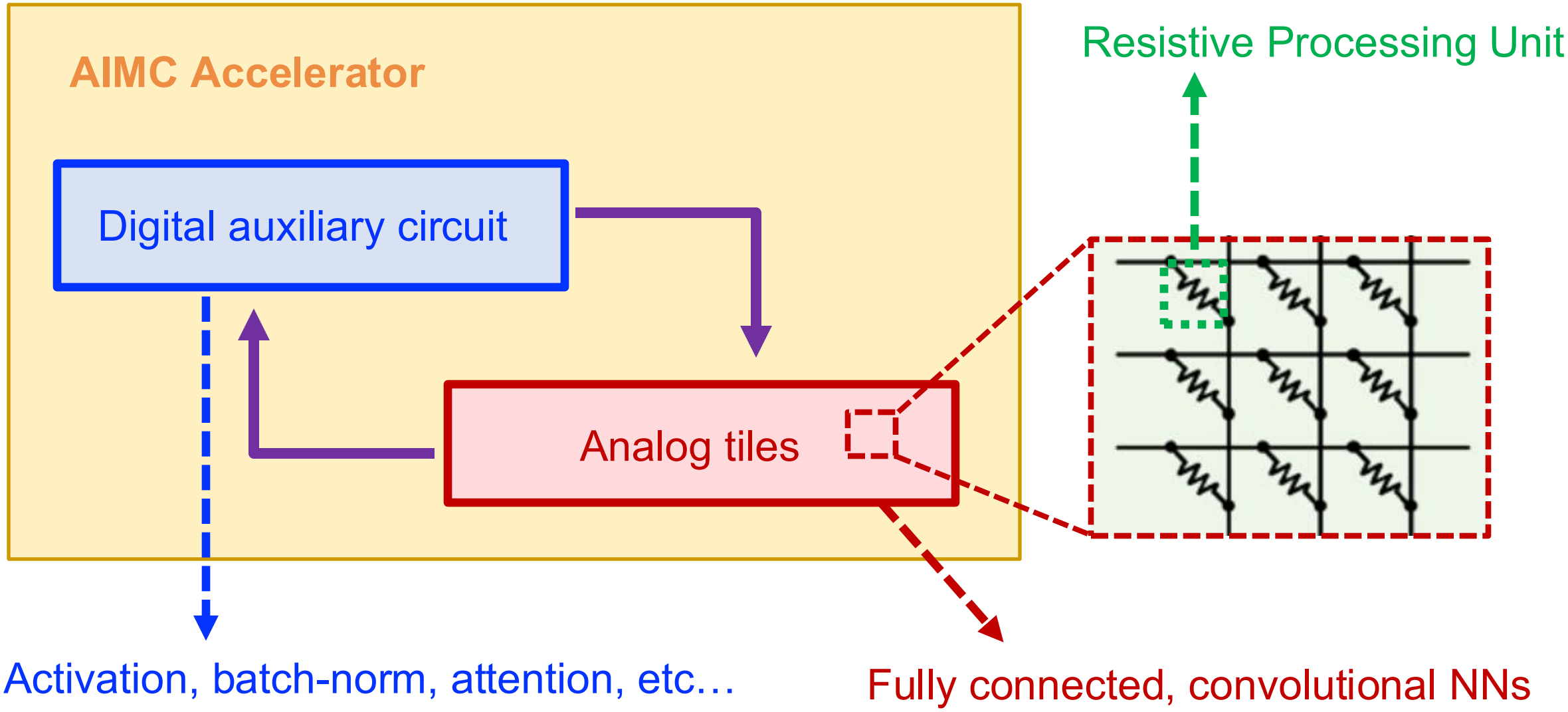
Motivation of analog in-memory computation (AIMC)

Matrix-Vector Multiplication (MVM) is the bottleneck of the DNN



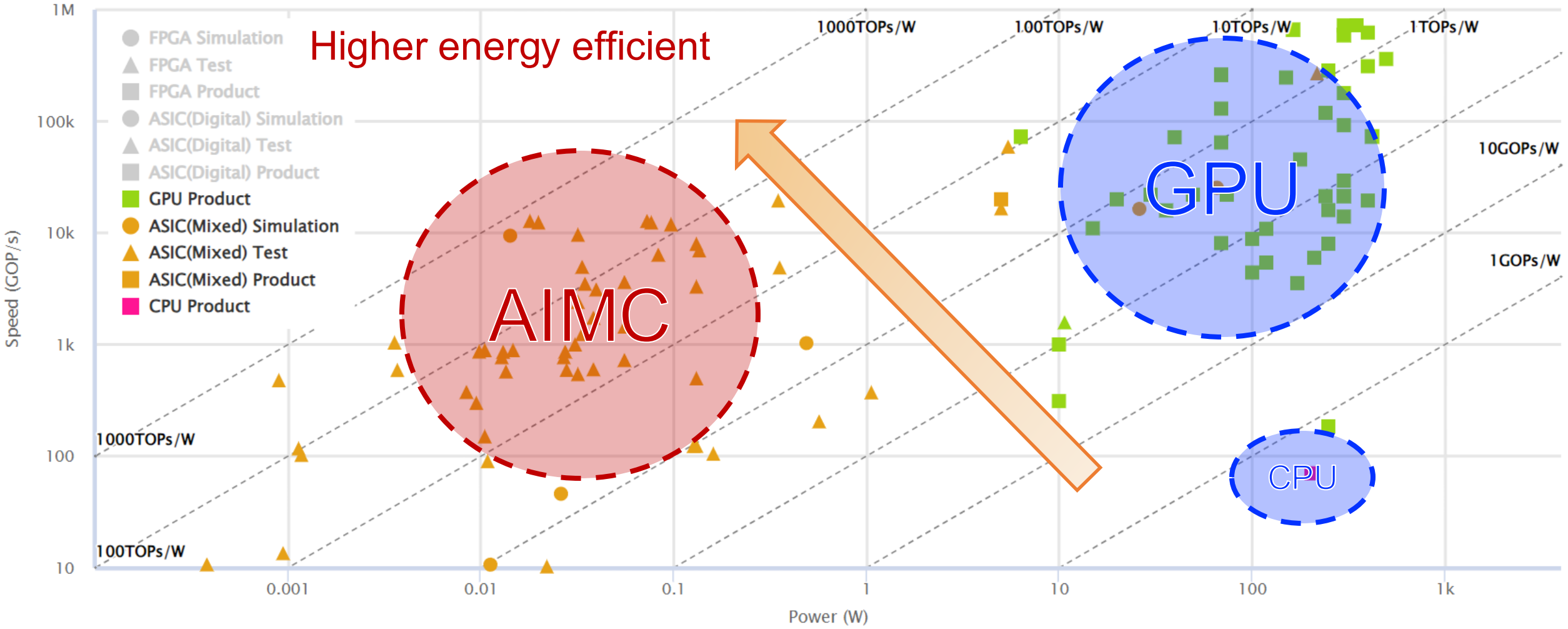
Z. Zhang, et al. "A universal RRAM-based DNN accelerator with programmable crossbars beyond MVM operator." **TCAD 41.7 (2021): 2094-2106**.

Overview of AIMC hardware



A Sebastian, et al. “Memory devices and applications for in-memory computing”. **Nature nanotechnology, 2020, 15(7): 529-544**
S Ambrogio, et al. “Equivalent-accuracy accelerated neural-network training using analogue memory”. **Nature, 2018, 558(7708): 60-67**.

Analog in-memory computing accelerates inference!



AIMC hardware: 10x-100x energy-efficient than GPU

K Guo, et al. "Neural Network Accelerator Comparison" **[Online]**. Available: https://nicsefc.ee.tsinghua.edu.cn/projects/neural-network-accelerator/

Significant real impact on inference cost

GPT-4o

GPT-4o is our most adv
Turbo with stronger vis
knowledge cutoff.

Learn about GPT-4o ↗

Model

gpt-4o

gpt-4o-2024-05-13

Vision pricing calculator

Set width Se

150 px by

Low resolution

← Post

William Wang
@WilliamWangNLP

Just another day, waking up to find out that the undergrad hired by your grad student generated a $20,000 OpenAI API bill in the last three days, without your authorization. 😅

10:42 AM · Jun 4, 2024 · **642.2K** Views

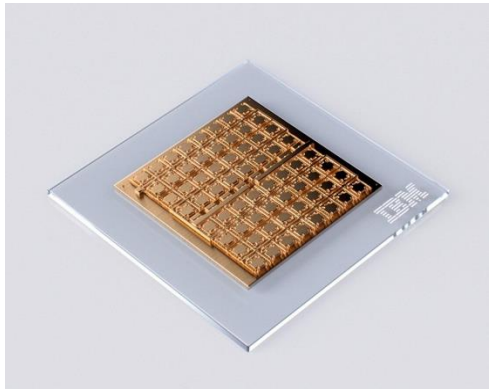
78 155 2.7K 398

Post your reply

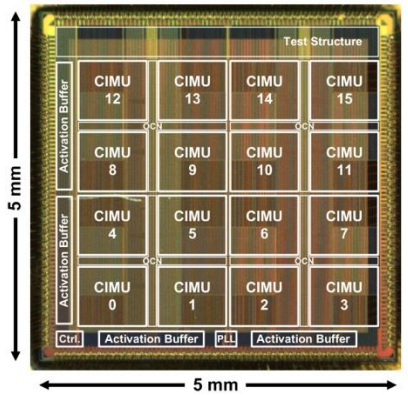
Reply

Images from Yin, Shukang, et al. "A survey on multimodal large language models." arXiv preprint arXiv:2306.13549 (2023), NVIDIA and https://medium.com/@ashwathkumarshettyr/multi-modal-large-language-models-1-introduction-f5e9bffc07df.

Recent AIMC chips



IBM HERMES chip
10.5 TOPS/W



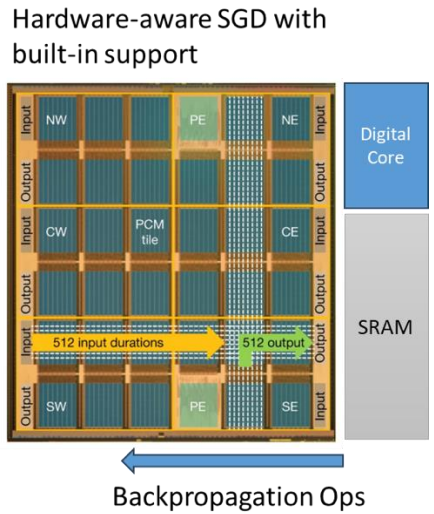
Enlarge AI chip
30 TOPS/W

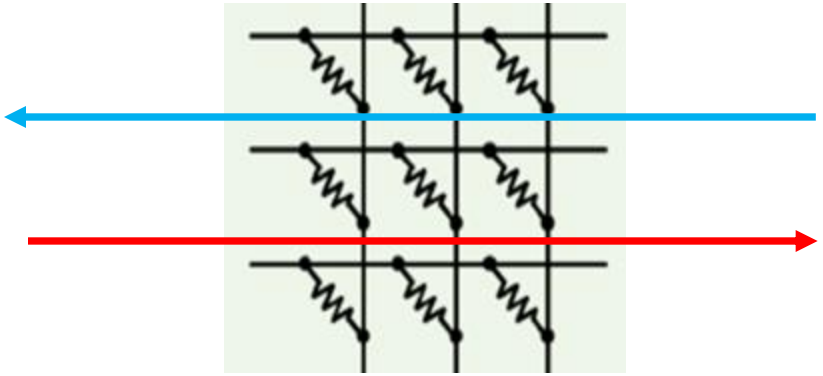
Mythic Analog Processor
6.3-8.3 TOPS/W

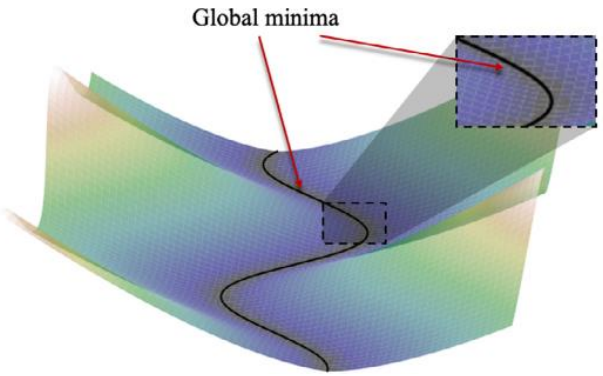
* TOPS/W = Tera Floating-point arithmetic Operations Per Second per Watt

M Le Gallo, et al. “A 64-core mixed-signal in-memory compute chip based on phase-change memory for deep neural network inference”**. Nature Electronics 6.9 (2023): 680-693.**
H Jia, et al. "15.1 a programmable neural-network inference accelerator based on scalable in-memory computing." IEEE International Solid-State Circuits Conference. Vol. 64, 2021.
M1076 Analog Matrix Processor. **https://mythic.ai/products/m1076-analog-matrix-processor/**

Scope of this lecture







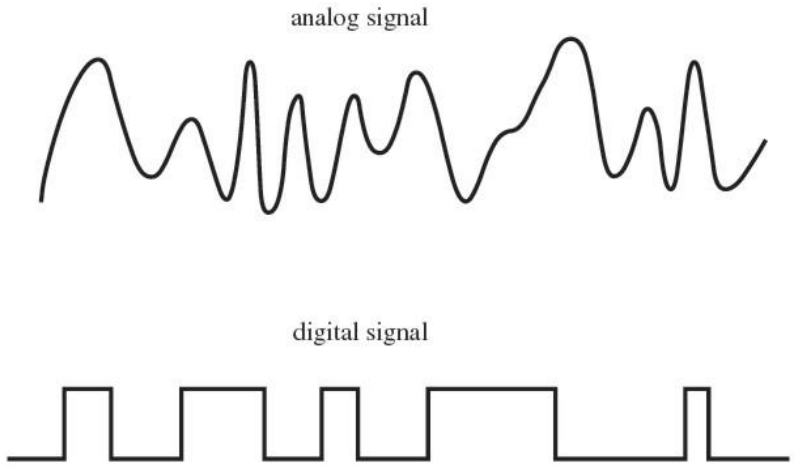
AIMC Circuit **Hardware property** **Analog AI**

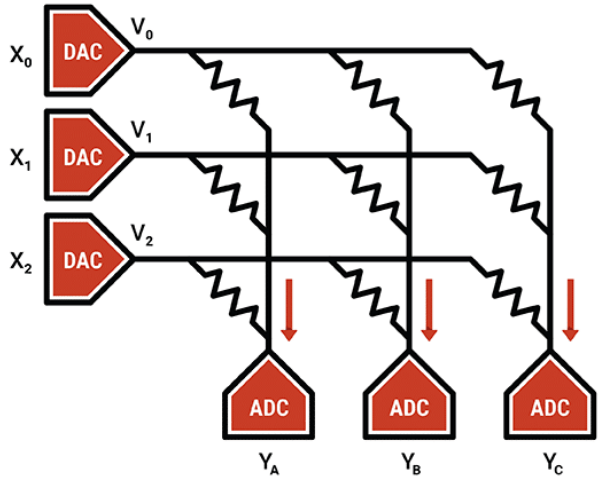
Hardware design

Algorithm design

Outline

- Background of Analog In-memory Computing
- Analog Inference: Hardware-aware Retraining





Resistive processing unit

Resistive processing unit (RPU): non-volatile electrical data storage unit

- Map the conductance (reciprocal of resistance) into the weight

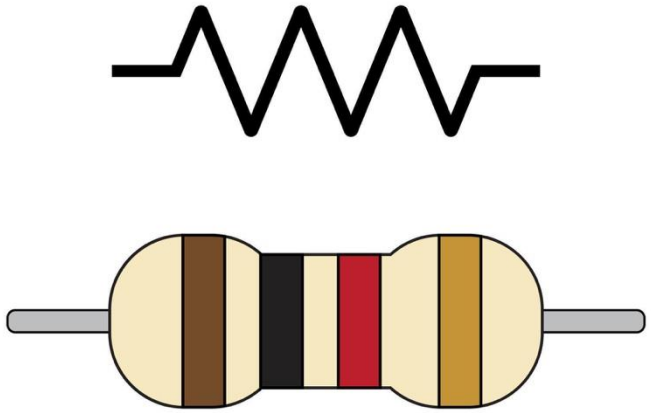
- The weight in RPU is an analog quantity (continuous quantity)

Analog Representation

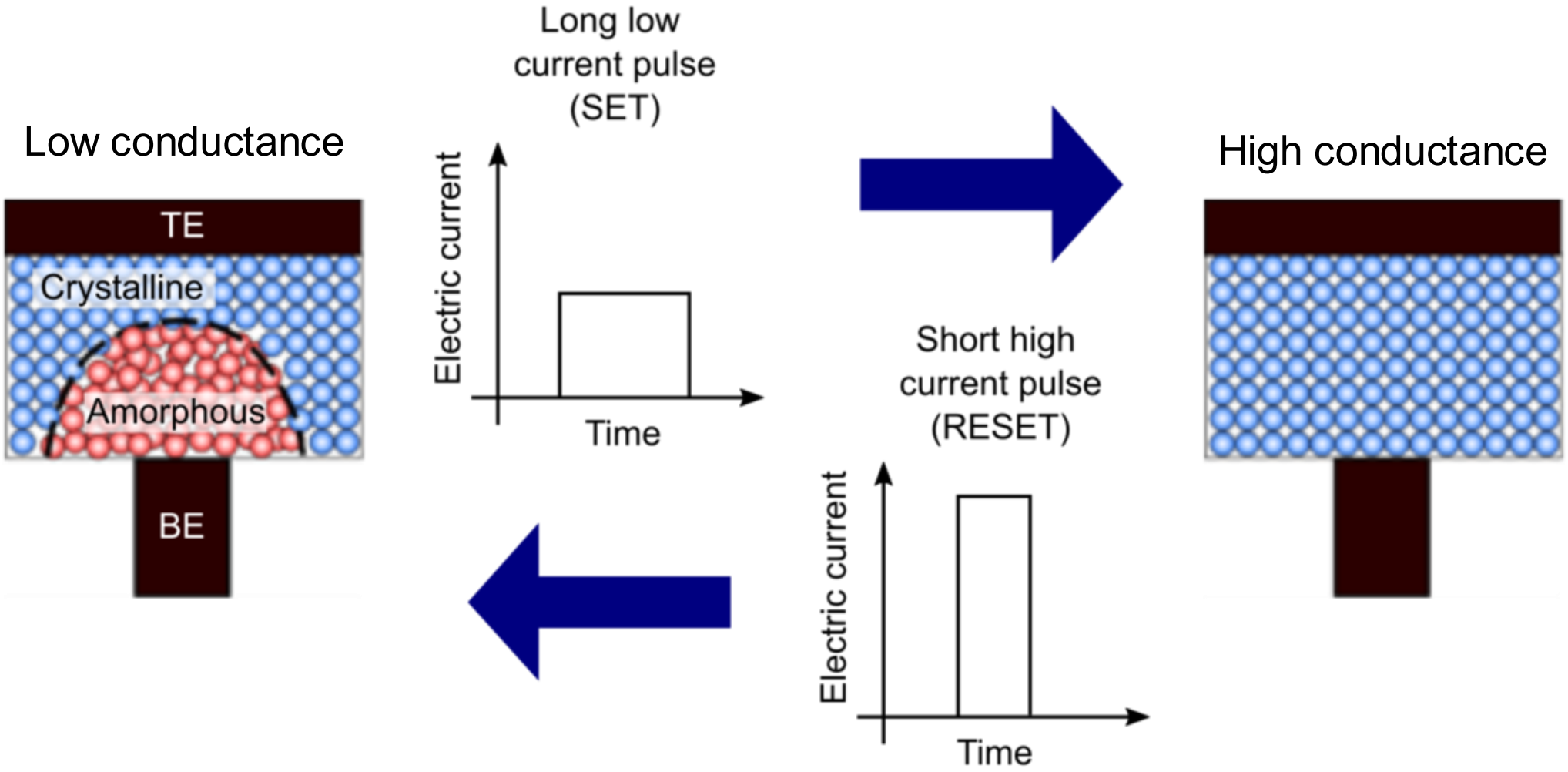
Conductance (nS)		Weight
10	->	0.1
20	->	0.2
30	->	0.3
40	->	0.4

Digital Representation

Code		Weight
001	->	0.1
010	->	0.2
011	->	0.3
100	->	0.4



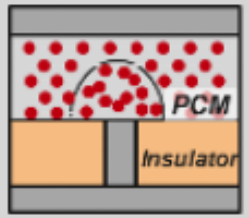
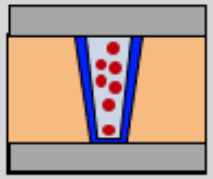
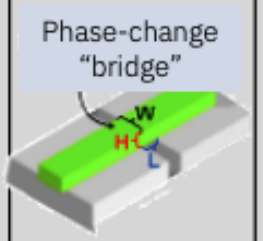
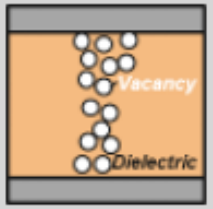
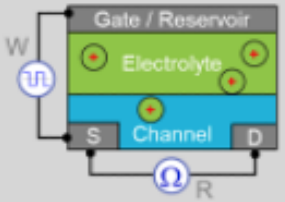
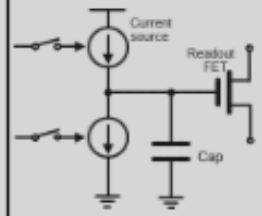
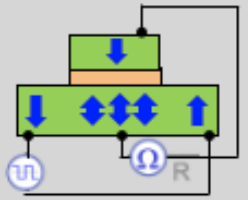
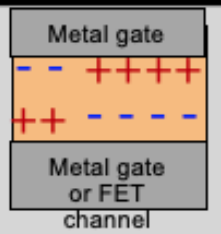
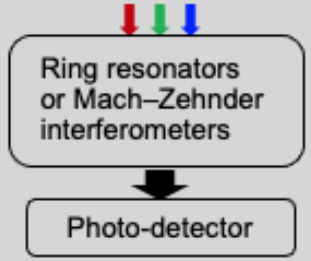
RPU example: Phase change memory



RPU can programmed by current pulses

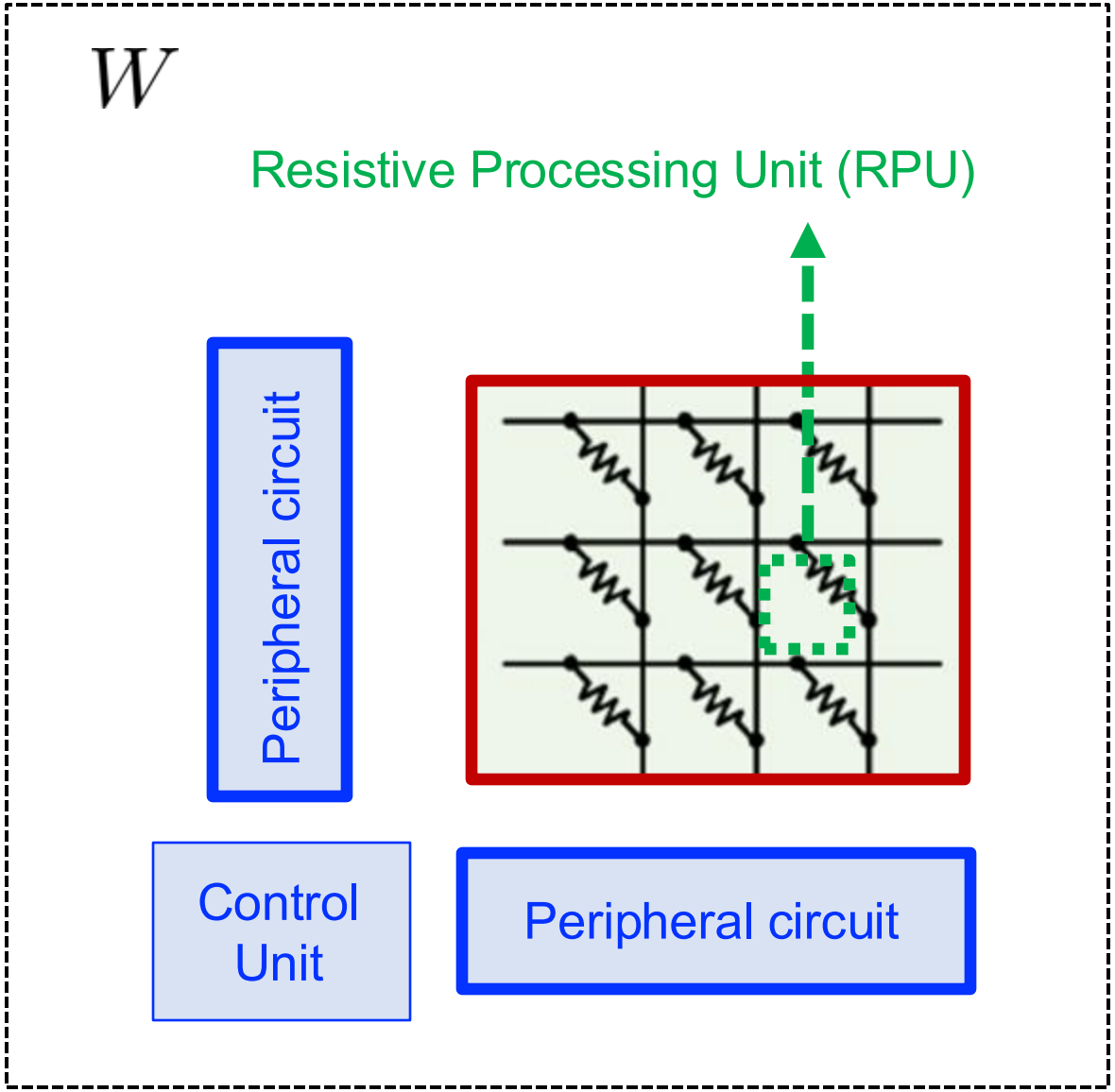
M Le Gallo, & A Sebastian. “An overview of phase-change memory device physics ”. **Journal of Physics D: Applied Physics, 53(21), 213002. 2020**

RPU candidates

Phase change memory (PCM) (crystalline/amorphous transition)			RRAM (filamentary)	ECRAM (Electrochem)	CMOS (capacitor)	MRAM & Ferroelectric (domain wall motion)		Photonics (transmission)
 mushroom	 confined surfactant					 magnetic	 FE	
+ Mature among storage-class memory			+ Bi-directional + Small	+ Symmetric + Low power	+Symmetric + Mature	+ Symmetric + Low power		+ Low power + Mature
- Abrupt reset - Asymmetry *(1-cell)* - Drift			- Stochastic - Asymmetry	- Open circuit voltage - Ion motion	- Cell size - Retention	- Domain wall control		- Cell size - Scalability (# of neurons)

Crossbar tile

- Arrange RPUs into a tile
- Peripheral circuits process the IO (analog-digital conversion)
- One crossbar tile stores one matrix W



Circuit class 101: Ohm's law and Kirchhoff law

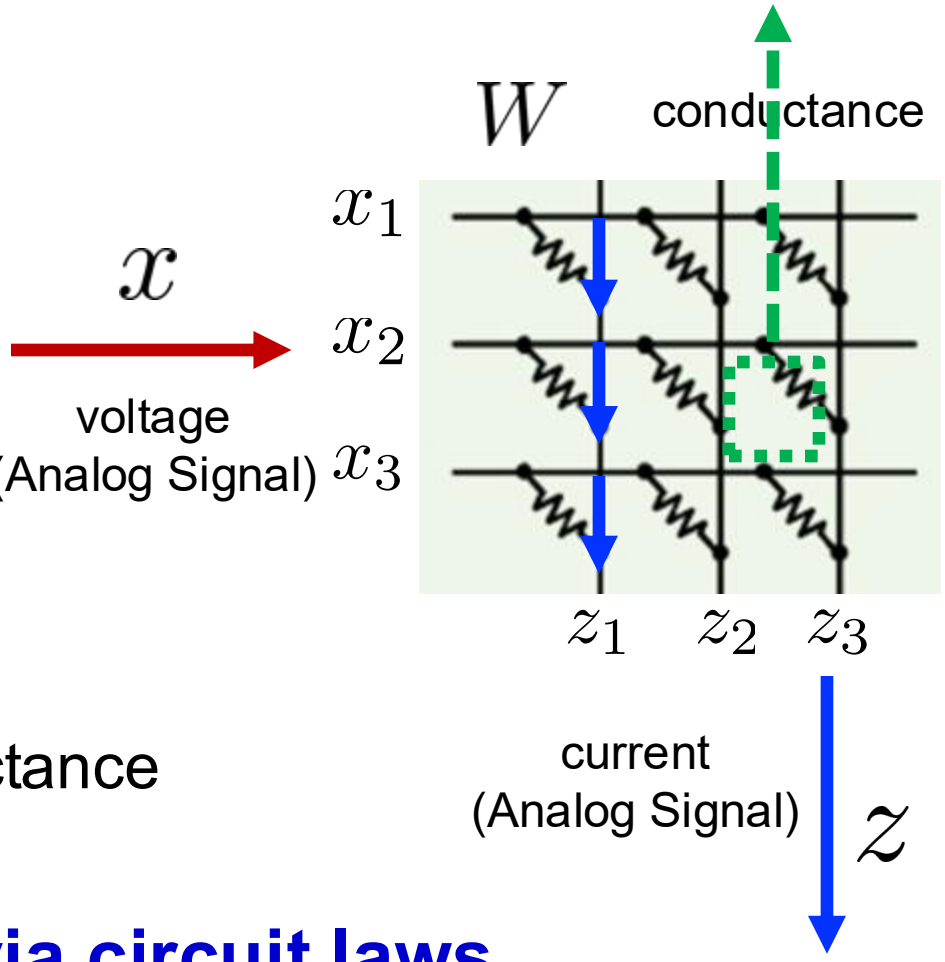
conductance

$$z = w \cdot x$$

current voltage

$$z_i = \underbrace{\sum_j \overbrace{W_{ij}x_j}^{\text{Ohm's law}}}_{\text{Kirchhoff's current law}}$$

Resistive Processing Unit (RPU)



x

voltage
(Analog Signal)

current
(Analog Signal)

z

- Basic circuit law: Ohm's and Kirchhoff's laws
- Data vector ***x*** is represented by the voltage
- Weight matrix ***W*** is represented by RPU's conductance

Matrix-vector multiplication done via circuit laws

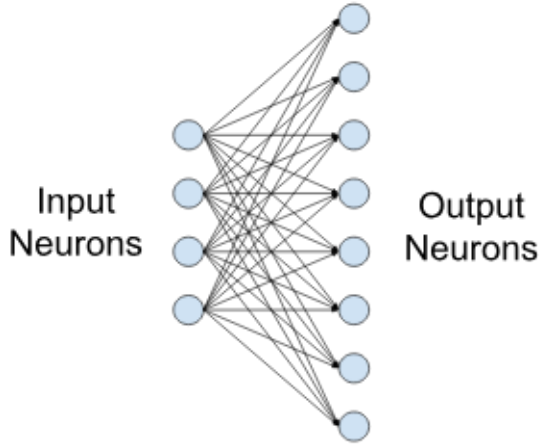
A Sebastian, et al. "Memory devices and applications for in-memory computing". **Nature nanotechnology, 2020, 15(7): 529-544**
S Ambrogio, et al. "Equivalent-accuracy accelerated neural-network training using analogue memory". **Nature, 2018, 558(7708): 60-67**.

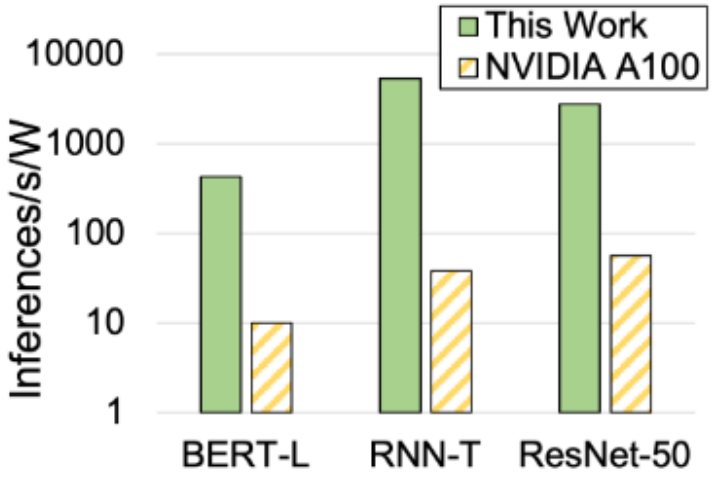
Linear layers on RPU array

- Linear layer is implemented by directly computing

  Output = Weight x Input + Bias

- Save heavy multiply–accumulate (MAC) computation





	Digital	Analog
MAC	$O(N^2)$	O(N)

Compare with NVIDIA A100:

- 100x-300x MVM energy-efficiency
- 40x-140x overall energy-efficiency

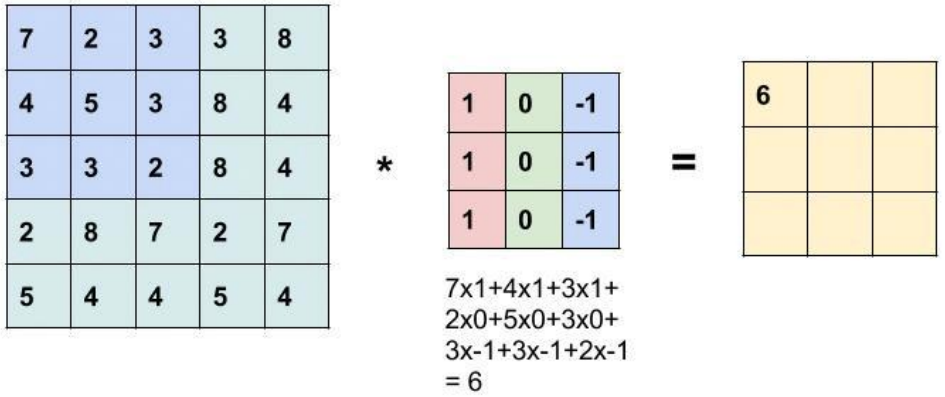
S Jain, et al. "A heterogeneous and programmable compute-in-memory accelerator architecture for analog-AI using dense 2-D mesh." **TVLSI 31.1 (2022): 114-127.**

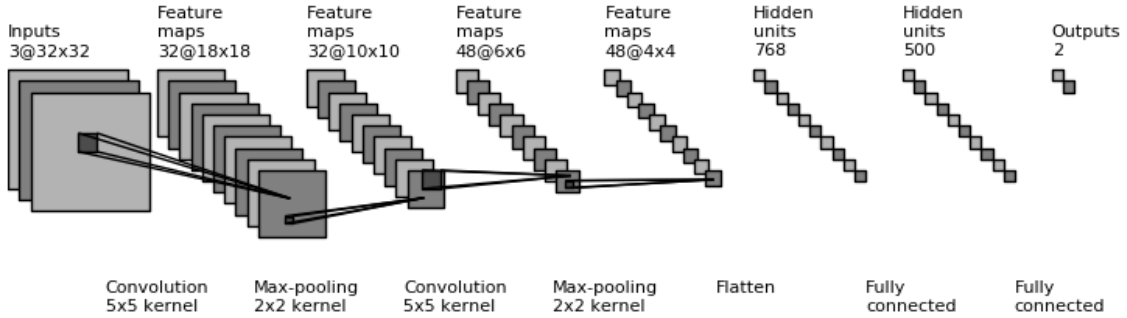
Convolutional layers on RPU array

Convolutional layer:

flatten each convolution kernel to a row of matrix

flatten image patches into a vector

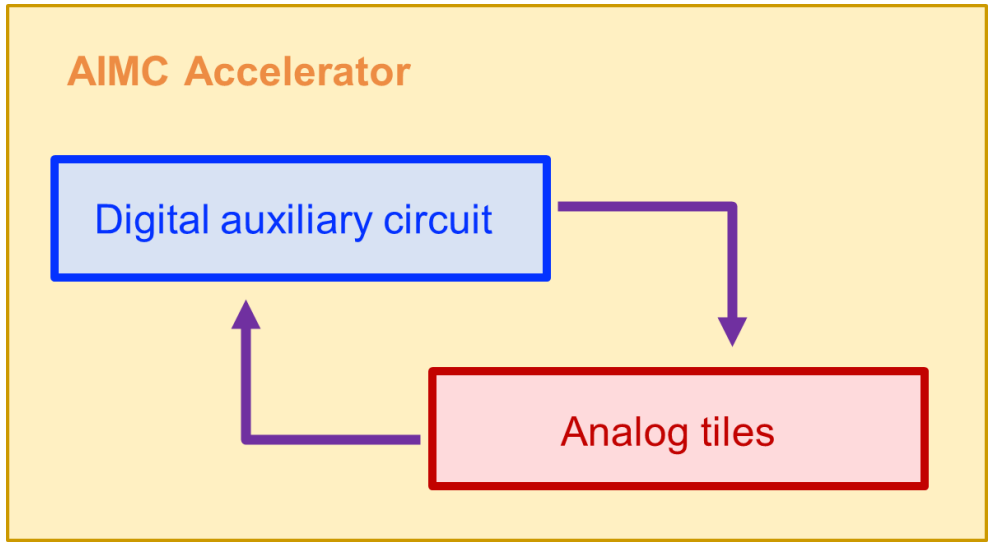


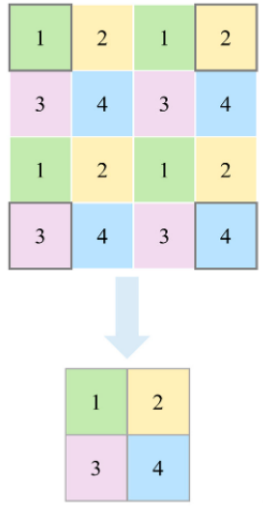


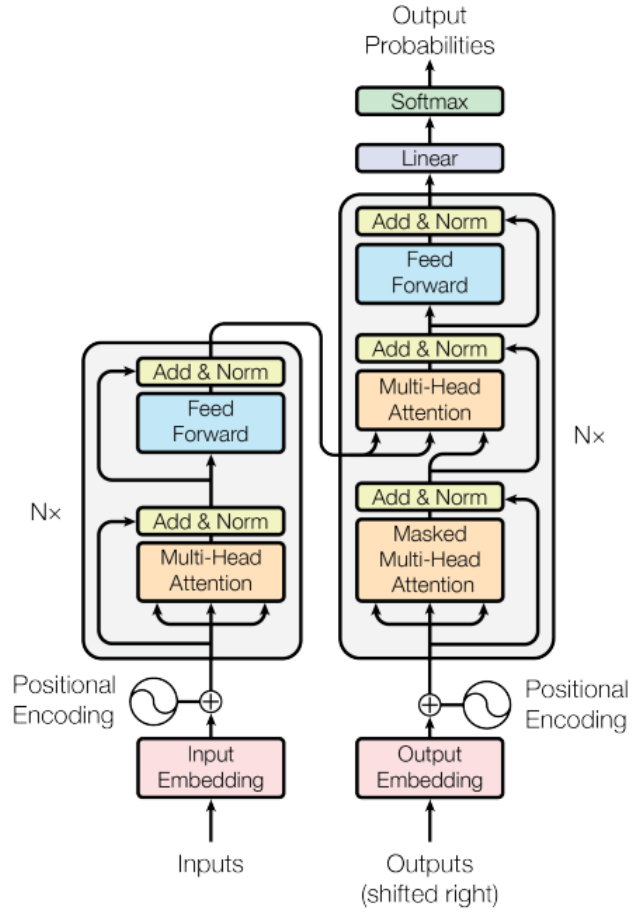
Non-linear operator on AIMC hardware

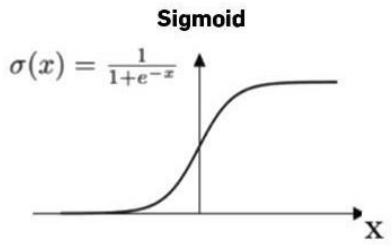
Non-linear operations are carried out in digital domain

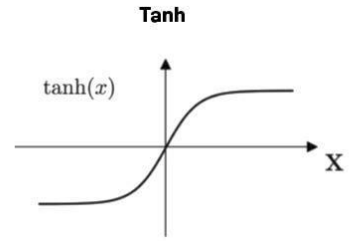
- Activation: Sigmoid, ReLU, LeakyReLU
- Down-sampling in CNN models
- Self-attention in Transform models

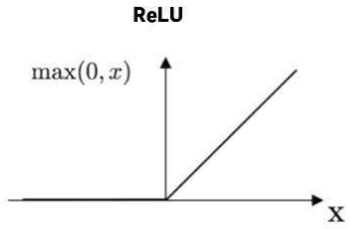




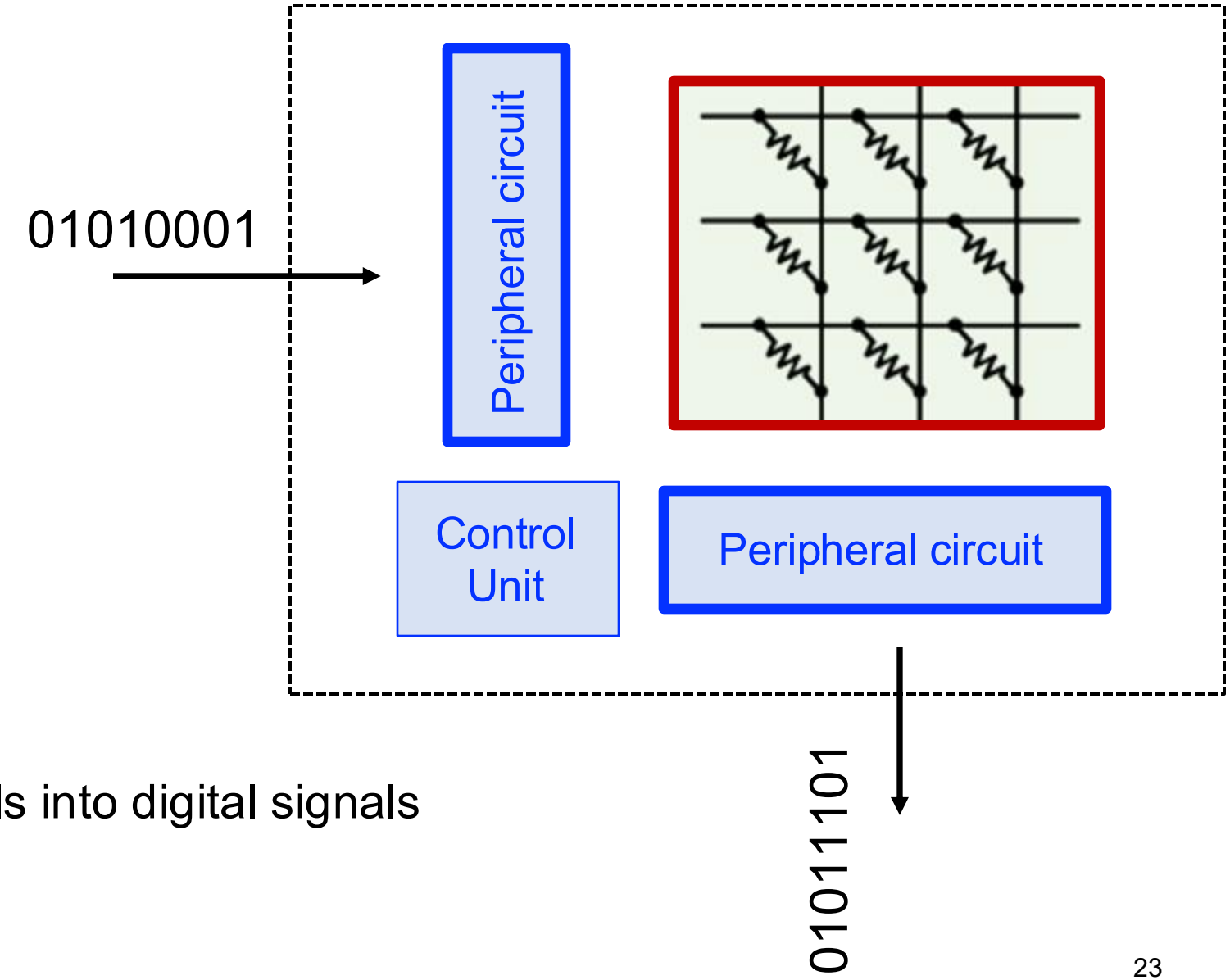








Analog-digital conversion



Peripheral circuit:

- Digital-Analog converter (DAC)
- Analog-Digital converter (ADC)

Quantize the current signals into digital signals

Weight programming

Pulse update: Send pulses to change conductance (thus weight)

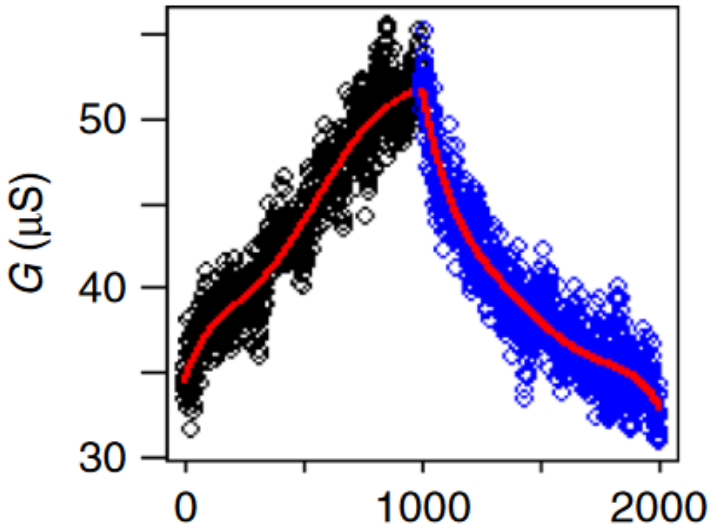
Program RPU for two purposes:

Inference: program pretrained model onto analog hardware

- Target weights are known

Training: keeping updating until model converge

- Target weights are unknown
- Adapt according to training objective



Different purposes leads to different update strategies

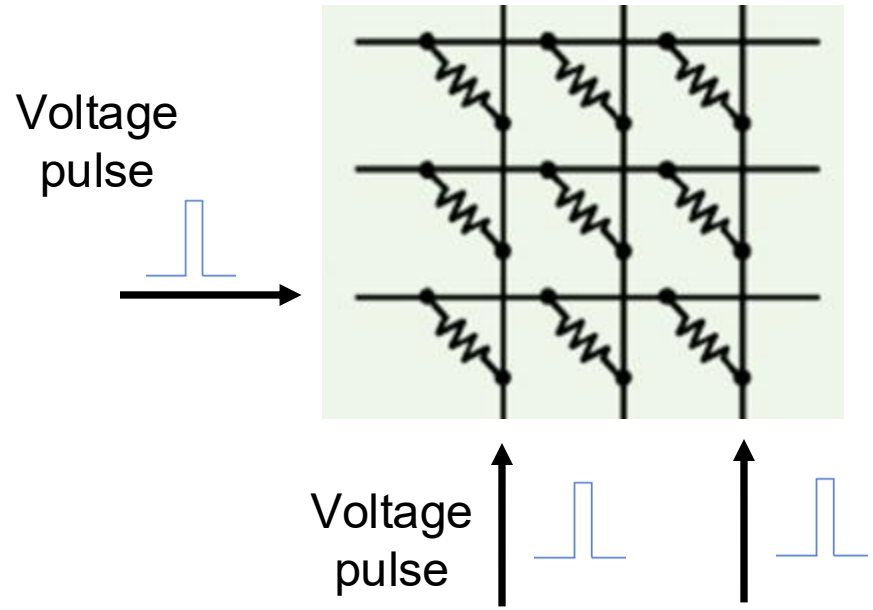
Weight programming for inference / training

Inference: expect to program the weights exactly

- Write-verification: ensure correct programming element-wise
- More accurate but circuits become more complicated

Training: only need to fulfill objective requirement

- Parallel update
- Idea: A pulse affects all elements simultaneously in the same row / column

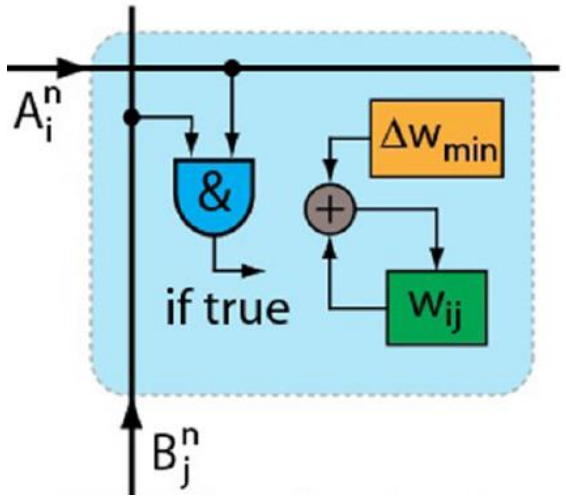


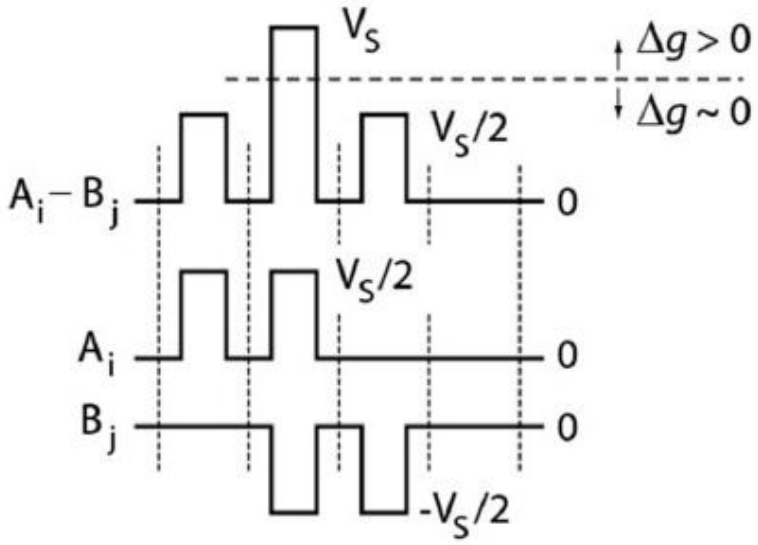
W Li, et al. "A 40-nm MLC-RRAM compute-in-memory macro with sparsity control, on-chip write-verify, and temperature-independent ADC references." **IEEE Journal of Solid-State Circuits 57.9 (2022): 2868-2877**.
Gokmen and Yurii. "Acceleration of deep neural network training with resistive cross-point devices: Design considerations." ***Frontiers in neuroscience*, 2016.**

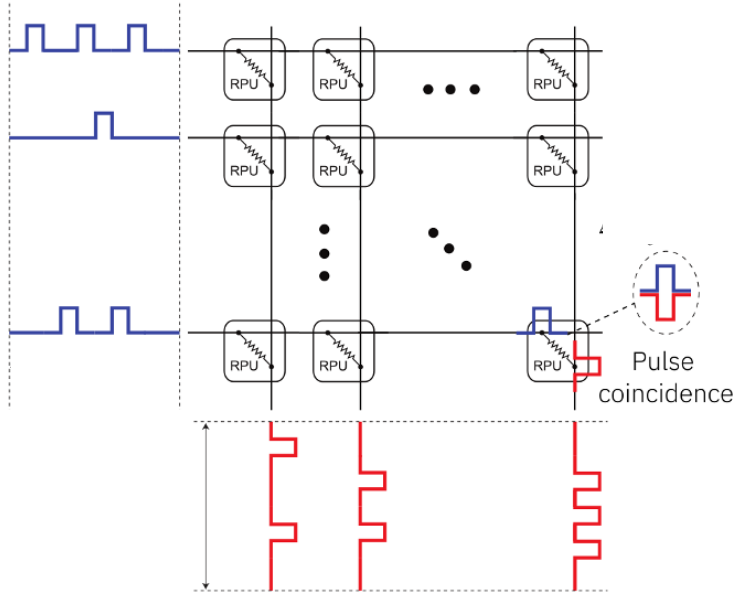
Parallel weight programming

Parallel update in crossbar tiles:

- Send pulses to both column & row
- Pulse coincident incurs weight update







Gokmen and Yurii. “Acceleration of deep neural network training with resistive cross-point devices: Design considerations.” ***Frontiers in neuroscience*, 2016.**

Weight copy

Direct copy:

Measure the conductance of each RPU

Parallel copy:

Taking MVM with one-hot vectors

3^{rd} *element*

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \cdot a_{11} + 0 \cdot a_{12} + 1 \cdot a_{13} + 0 \cdot a_{14} \\ 0 \cdot a_{21} + 0 \cdot a_{22} + 1 \cdot a_{23} + 0 \cdot a_{24} \\ 0 \cdot a_{31} + 0 \cdot a_{32} + 1 \cdot a_{33} + 0 \cdot a_{34} \\ 0 \cdot a_{41} + 0 \cdot a_{42} + 1 \cdot a_{43} + 0 \cdot a_{44} \end{bmatrix} = \begin{bmatrix} 1 \cdot a_{13} \\ 1 \cdot a_{23} \\ 1 \cdot a_{33} \\ 1 \cdot a_{43} \end{bmatrix} = \begin{bmatrix} a_{13} \\ a_{23} \\ a_{33} \\ a_{43} \end{bmatrix}$$

3^{rd} *column selected*

Parallel copy is more efficient if only a few columns are needed

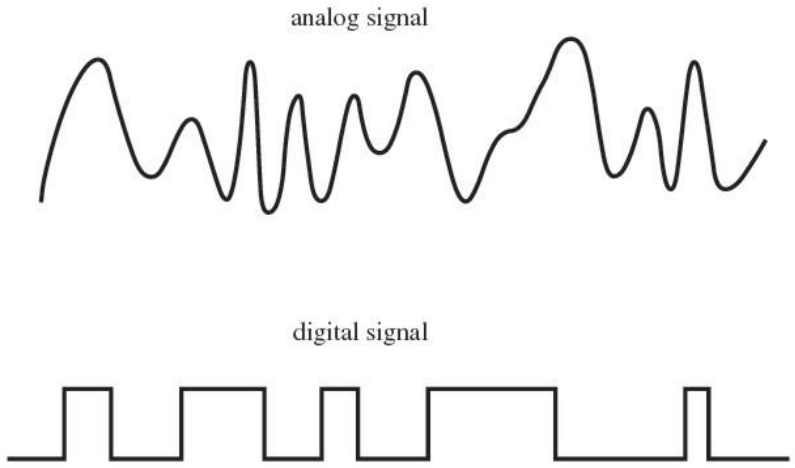
Takeaway

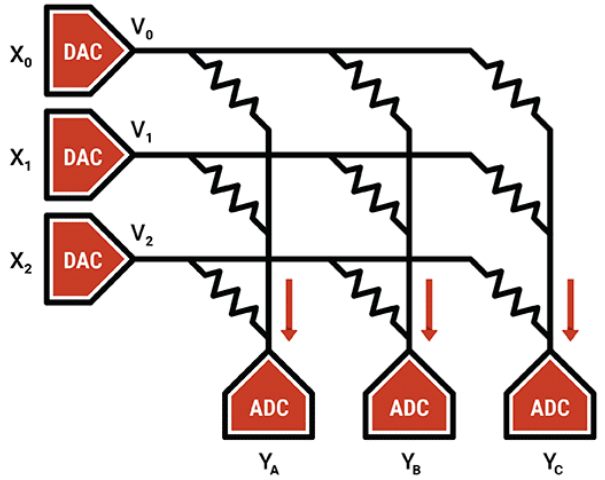
Basic operations on AIMC hardware:

- **Resistive processing unit (RPU)** leverages conductance to represent value
- **Matrix-vector multiplication (MVM):** Ohm's and Kirchhoff's laws
- Programming: **Pulse update**
  - Element-wise write & verification
  - Parallel programming
- Weight copy:
  - Measure conductance directly
  - Copying by MVM with one-hot vectors

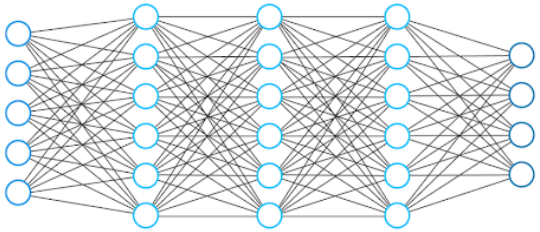
Outline

- Background of Analog In-memory Computing
- Analog Inference & Hardware-aware Retraining





Inference on AIMC hardware



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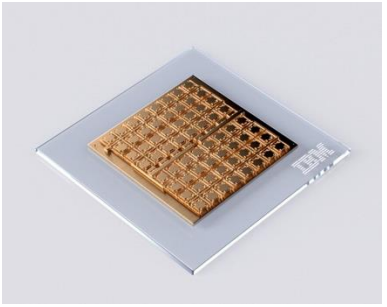
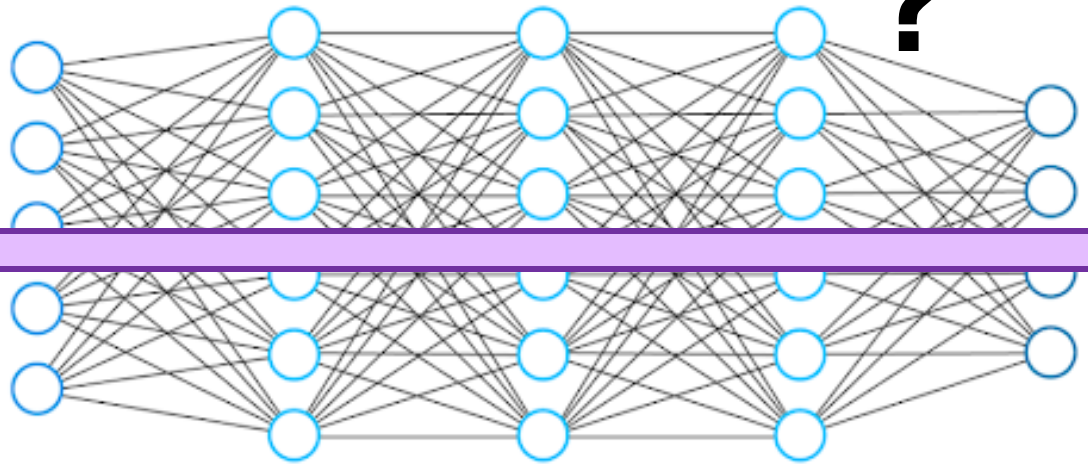


Pretrain a Model on GPUs

Pretrained Model

Program on AIMC hardware

?

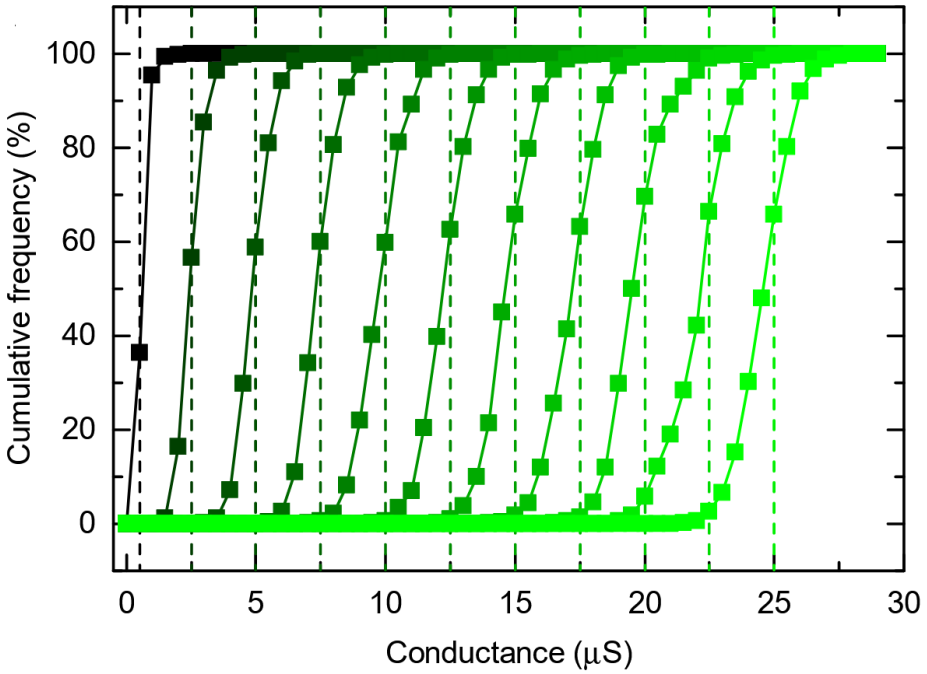


CAT

Both programming and inference can be problematic

Hardware imperfection I: Programming error

- The conductance values programmed on the hardware have a certain error
- Fig: Cumulative distributions of 11 representative iteratively programmed conductance levels

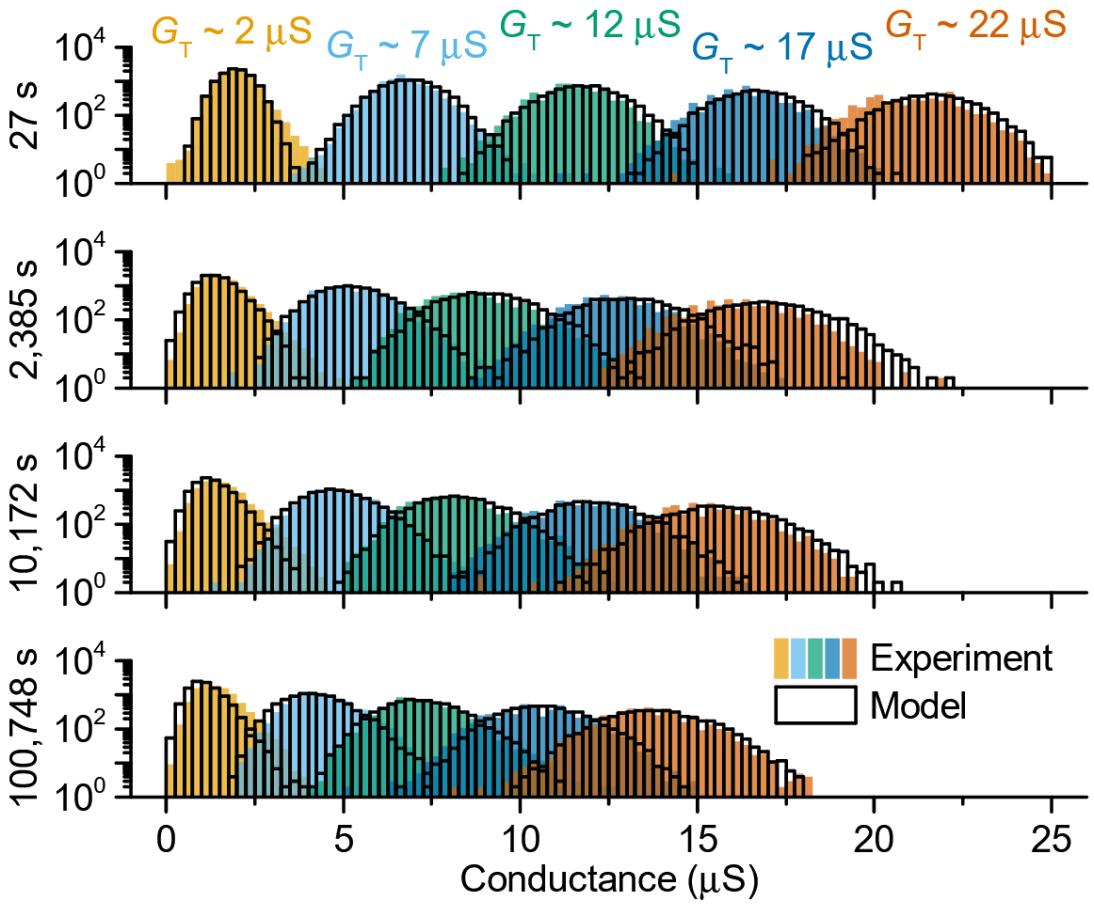


The programming processing is inaccurate

V Joshi, et al. "Accurate deep neural network inference using computational phase-change memory." **Nature communications 11.1 (2020): 2473.**

Hardware imperfection II: Weight drift

- Programmed weights drift over time
- PCM suffers from structural relaxation of the amorphous phase



V Joshi, et al. "Accurate deep neural network inference using computational phase-change memory." **Nature communications 11.1 (2020): 2473.**

Hardware imperfection II: Weight drift (con't)

Direct mapping	Test Error in %		
DNN	FP_{32}	1 hour	1 year
ResNet-32 CF10	5.80	11.68 0.16	15.21 0.40
WideRN-16 CF100	20.00	26.02 0.10	29.42 0.28
ResNet-18† ImNet	30.50	47.91 0.21	55.00 0.41
ResNet-50† ImNet	23.87	32.51 0.09	38.18 0.31
DenseNet-121† ImNet	25.57	47.38 0.24	59.81 0.71
WideRN-50† ImNet	21.53	40.07 0.13	45.45 0.30
BERT-base MRPC	14.60	15.99 0.23	18.36 0.30
Albert-base MRPC	15.08	31.50 0.10	31.62 0.03
Speech□ SWB300	14.05	16.11 0.02	16.32 0.02
LSTM PTB	72.90	73.16 0.01	73.28 0.01
RNN-T SWB300	11.80	28.16 0.10	29.08 0.16

Weight drift incurs significant accuracy degradation

V Joshi, et al. "Accurate deep neural network inference using computational phase-change memory." **Nature communications 11.1 (2020): 2473.**

Analog hardware-aware retraining

Idea: retrain with considering all the possible noise

min Loss(W)
over **Model weight (W)**

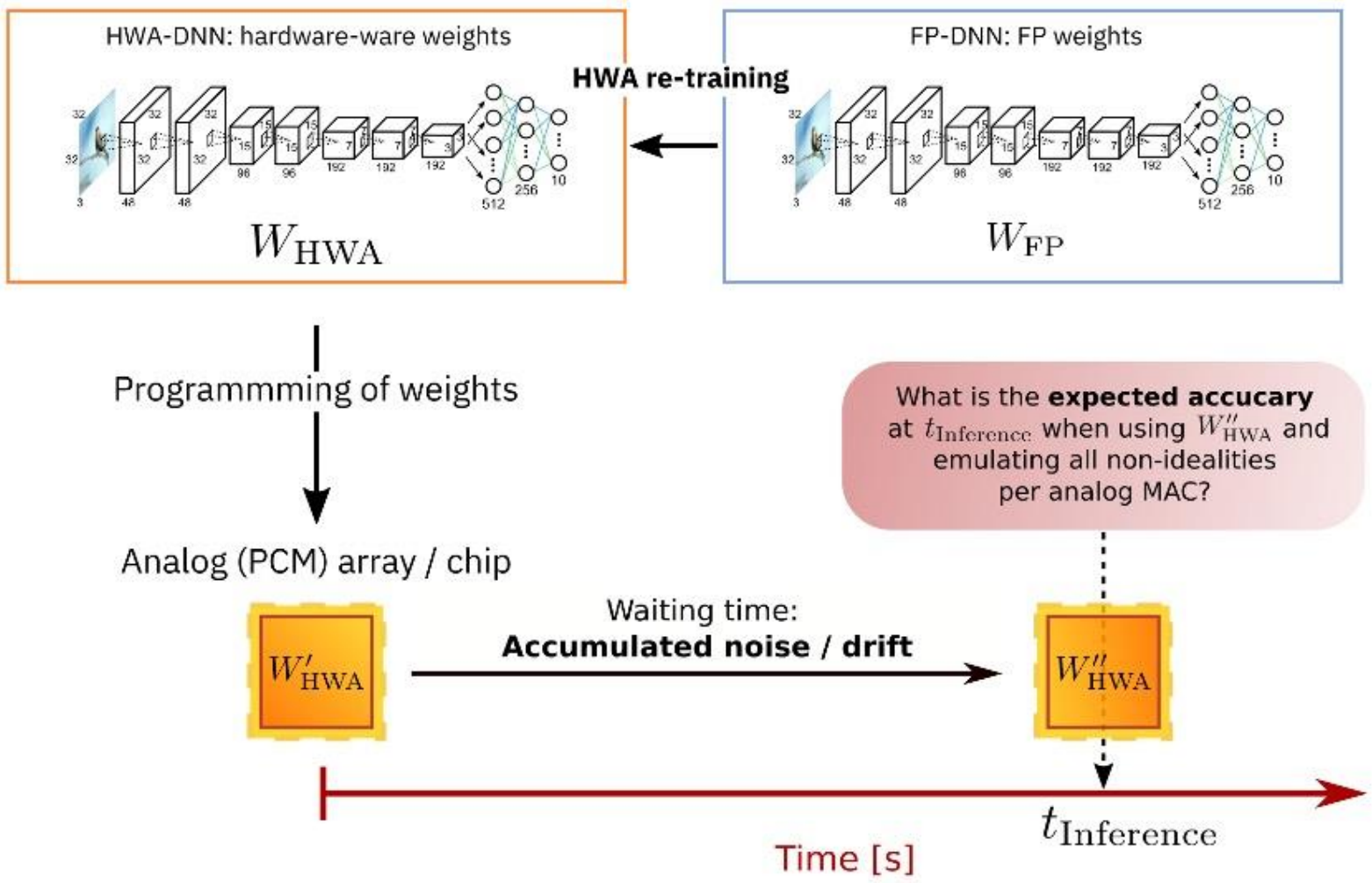
Original objective

min Loss(W + possible drift)
over Model Weight (W)

hardware-aware Retraining
(HWA) objective

V Joshi, et al. "Accurate deep neural network inference using computational phase-change memory." **Nature communications 11.1 (2020): 2473.**
MJ, Rasch, et al. "Hardware-aware training for large-scale and diverse deep learning inference workloads using in-memory computing-based accelerators." **Nature communications 14.1 (2023): 5282.**

Analog hardware-aware retraining (con't)



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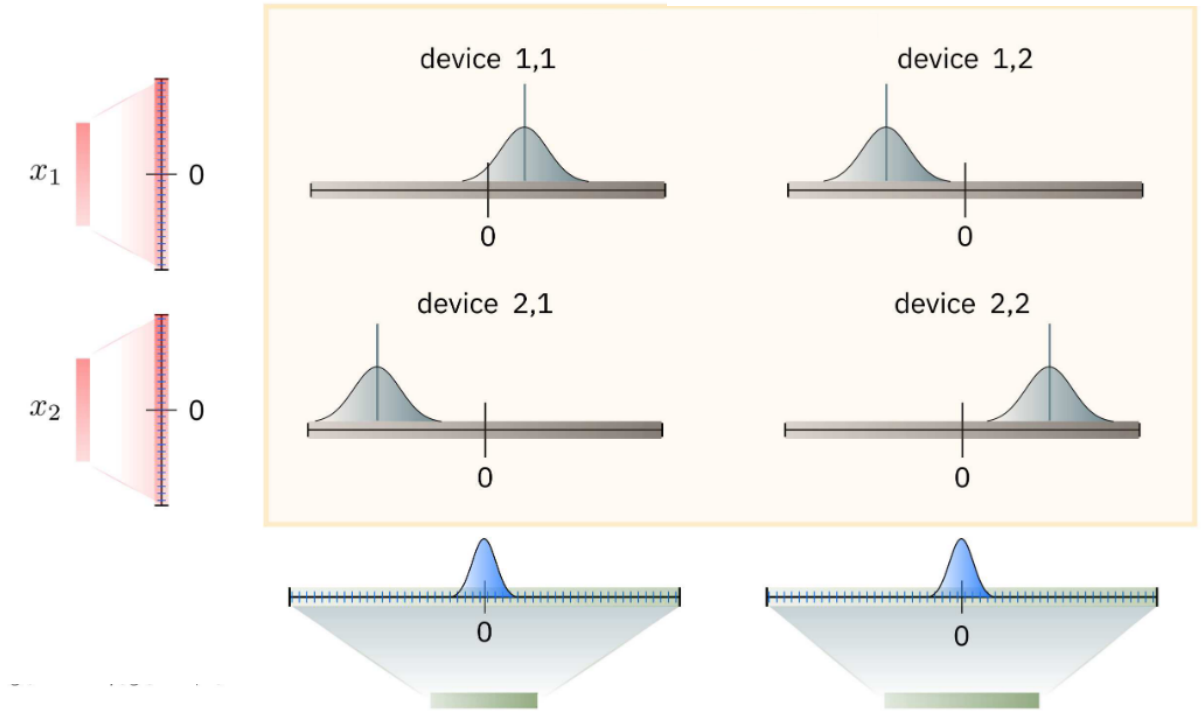
Hardware imperfection III: Quantization & read noise

- Weights suffers from instantaneous fluctuations due to the intrinsic noise

Output = Quad((Weight + noise) x Input + Bias)

Similar signals are quantized to same level

$$z = \mathrm{Quad}((W + \varepsilon)x)$$



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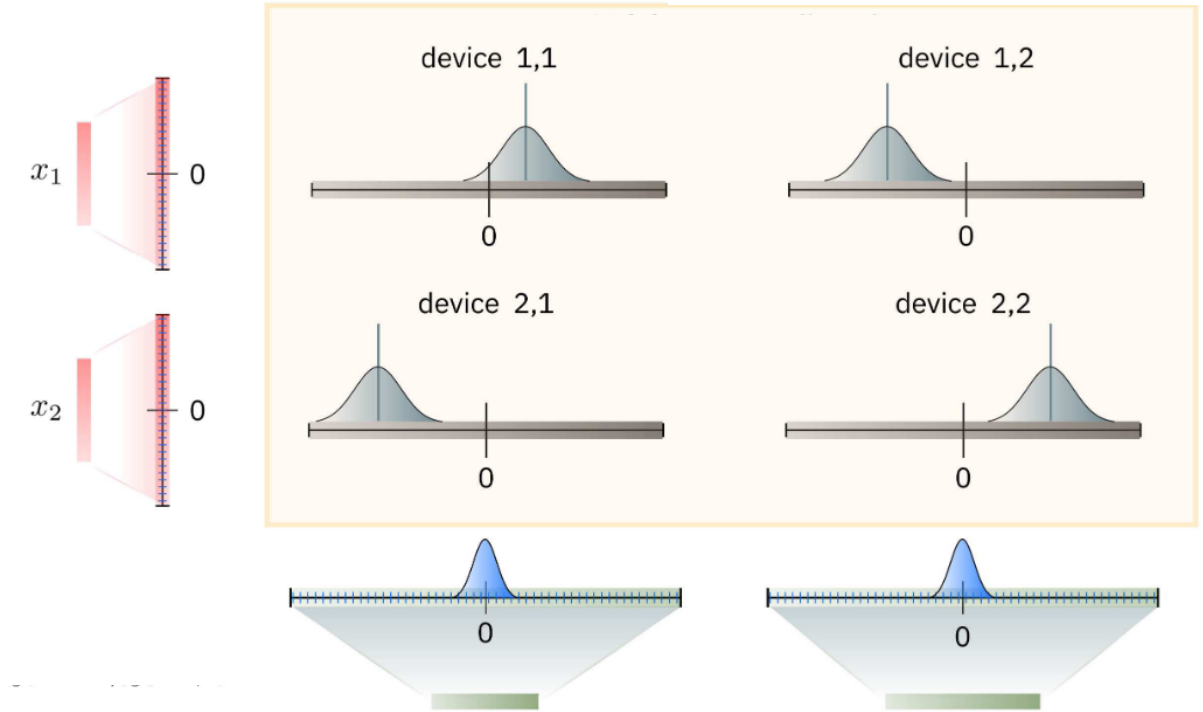
Hardware imperfection III: Quantization & read noise

- Solution: scale the input before MVM

$$\tilde{x} = \frac{x}{\alpha} \quad \tilde{z} = \mathrm{Quad}((W + \varepsilon)\tilde{x}) \quad z = \alpha\tilde{z}$$

- Equivalently

$$z = \alpha \cdot \mathrm{Quad}((W + \epsilon) \cdot \frac{x}{\alpha})$$



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